



OSCILLATIONS, WAVES AND OPTICS

PHYSICS FOR CHEMISTRY STUDENTS

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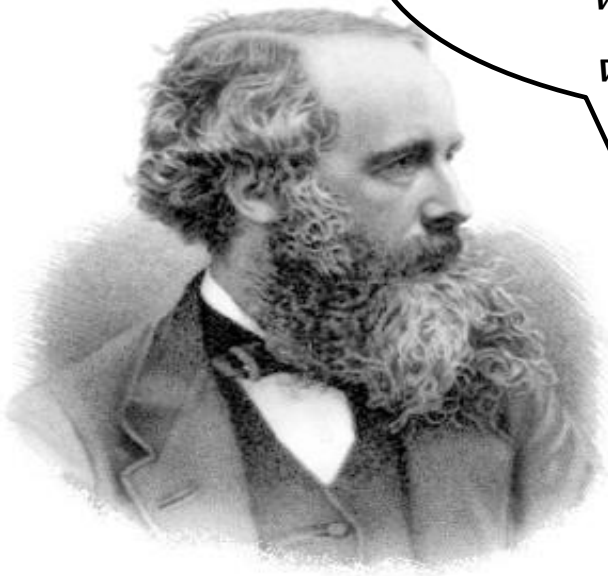
REVIEW-EM WAVE EQ.

$$\nabla \times \mathbf{E}(\mathbf{r}, t) = -\frac{\partial \mathbf{B}(\mathbf{r}, t)}{\partial t}$$

$$\nabla \times \mathbf{H}(\mathbf{r}, t) = \frac{\partial \mathbf{D}(\mathbf{r}, t)}{\partial t} + \mathbf{j}(\mathbf{r}, t)$$

$$\nabla \cdot \mathbf{D}(\mathbf{r}, t) = \rho(\mathbf{r}, t)$$

$$\nabla \cdot \mathbf{B}(\mathbf{r}, t) = 0$$



$$\frac{\partial^2}{\partial t^2} \mathbf{E}(\mathbf{r}, t) = \frac{1}{\epsilon_0 \mu_0} \nabla^2 \mathbf{E}(\mathbf{r}, t)$$

$$v_p = \sqrt{\frac{1}{\epsilon_0 \mu_0}} = c$$

REVIEW

$$\frac{\partial^2 u}{\partial t^2} = v_p^2 \frac{\partial^2 u}{\partial x^2}$$

General solution of the wave equation

$$u(x, t) = g(x + v_p t) + f(x - v_p t)$$



REVIEW-EM WAVE EQ.

Wave equations

$$\frac{\partial^2}{\partial t^2} \mathbf{B}(r, t) = \frac{1}{\epsilon_0 \mu_0} \nabla^2 \mathbf{B}(r, t)$$

$$\frac{\partial^2}{\partial t^2} \mathbf{E}(r, t) = \frac{1}{\epsilon_0 \mu_0} \nabla^2 \mathbf{E}(r, t)$$

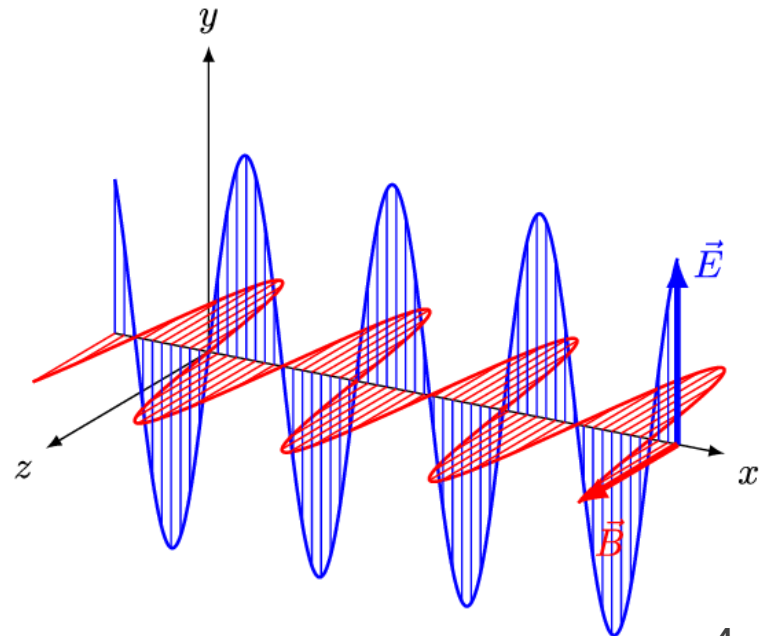
$$\mathbf{B}(x, t) \times \mathbf{E} = \vec{k}$$

Free Space Solutions (harmonic):

$$\mathbf{E}(r, t) = \mathbf{E}_0 e^{i\mathbf{k}r - i\omega t}$$

$$\mathbf{B}(r, t) = \mathbf{B}_0 e^{i\mathbf{k}r - i\omega t}$$

$$v_p = \frac{\omega}{k} = \sqrt{\frac{1}{\epsilon_0 \mu_0}}$$



Transverse wave

REVIEW-EM WAVE EQ.

Wave equations

$$\frac{\partial^2}{\partial t^2} \mathbf{B}(r, t) = \frac{1}{\epsilon_0 \mu_0} \nabla^2 \mathbf{B}(r, t)$$

$$\frac{\partial^2}{\partial t^2} \mathbf{E}(r, t) = \frac{1}{\epsilon_0 \mu_0} \nabla^2 \mathbf{E}(r, t)$$

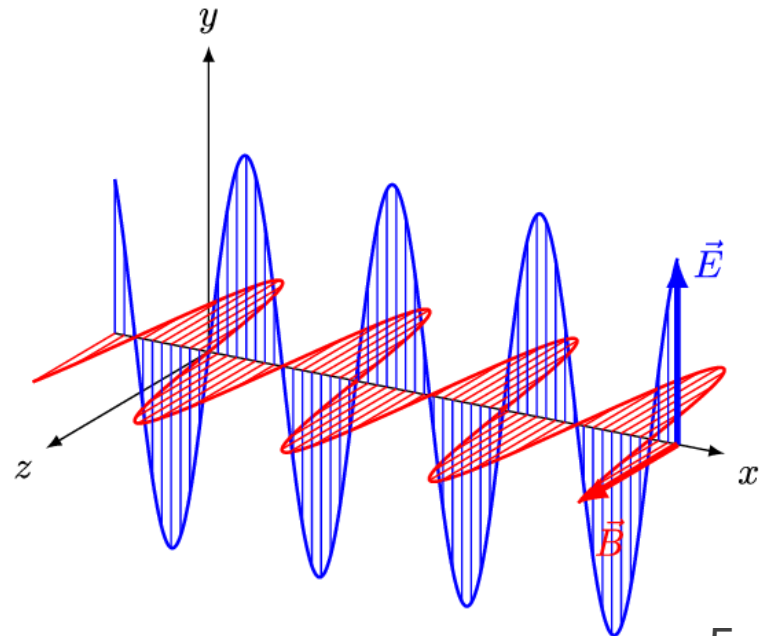
$$\mathbf{B}(x, t) \times \mathbf{E} = \vec{k}$$

Solutions:

$$\mathbf{E}(r, t) = \mathbf{E}_0 \sin(kr - \omega t)$$

$$\mathbf{B}(r, t) = \mathbf{B}_0 \sin(kr - \omega t)$$

$$v_p = \frac{\omega}{k} = \sqrt{\frac{1}{\epsilon_0 \mu_0}}$$



Transverse wave

REVIEW-EM POWER DENSITY

Solutions:

$$\mathbf{E}(r, t) = \mathbf{E}_0 \sin(kr - \omega t) \quad \mathbf{B}(r, t) = \mathbf{B}_0 \sin(kr - \omega t)$$

$$\nabla \times \mathbf{E}(x, t) = -\frac{\partial \mathbf{B}(x, t)}{\partial t} \quad \Rightarrow \quad ikE(x, t) = i\omega B(x, t)$$

$$\langle \mathbf{S} \rangle = \left\langle \frac{1}{\mu_0} \mathbf{E} \times \mathbf{B} \right\rangle = \frac{1}{\mu_0} |E_0 B_0| \langle \sin^2 \omega t \rangle = \frac{1}{2} \frac{1}{c\mu_0} (E_0)^2 = \frac{1}{2} \varepsilon_0 c (E_0)^2$$

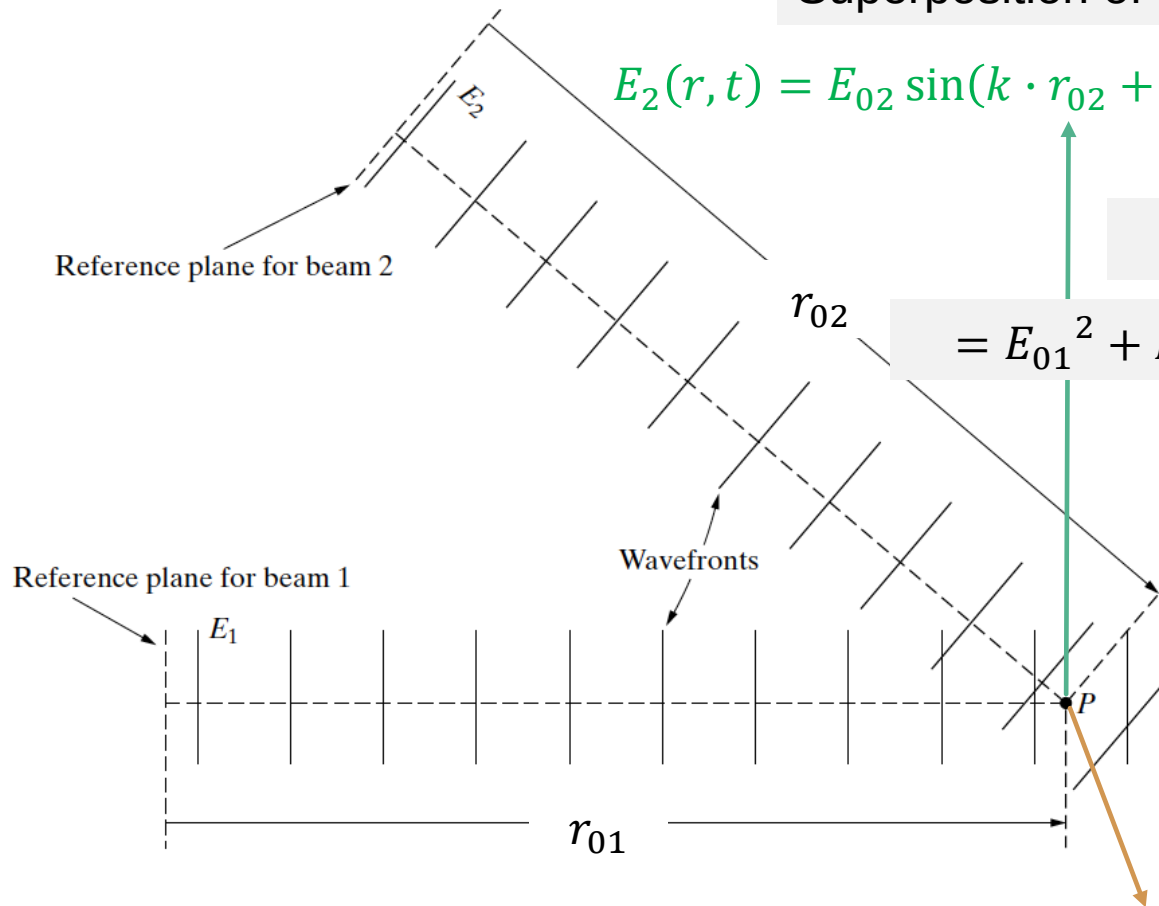
REVIEW-SUPERPOSITION (I)

Superposition of waves of the same frequency

$$E_2(r, t) = E_{02} \sin(k \cdot r_{02} + \omega t + \varphi_{02}) = E_{02} \sin(\omega t + \alpha_{02})$$

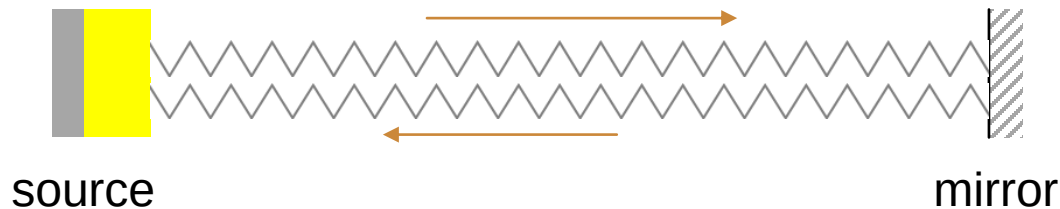
$$I \propto E_{00}^2$$

$$= E_{01}^2 + E_{02}^2 + 2E_{01}E_{02} \cos(\alpha_{01} - \alpha_{02})$$



$$E_1(r, t) = E_{01} \sin(k \cdot r_{01} + \omega t + \varphi_{01}) = E_{01} \sin(\omega t + \alpha_{01})$$

REVIEW-SUPERPOSITION (II)

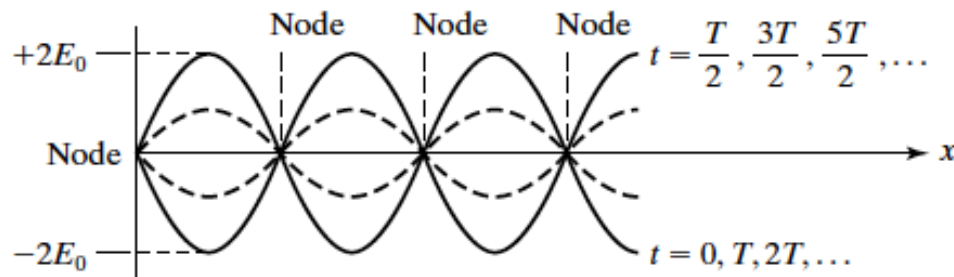


Forward: $E_1(\mathbf{r}, t) = E_0 \sin(kx - \omega t)$

backward: $E_2(\mathbf{r}, t) = E_0 \sin(kx + \omega t)$

$$E_r(\mathbf{r}, t) = E_1 + E_2 = 2E_0 \sin(kx) \cos(\omega t)$$

Nodes $kx = m\pi$



$$x = m \frac{\pi}{k} = m \frac{\lambda}{2} = 0, \frac{\lambda}{2}, \lambda, \dots$$

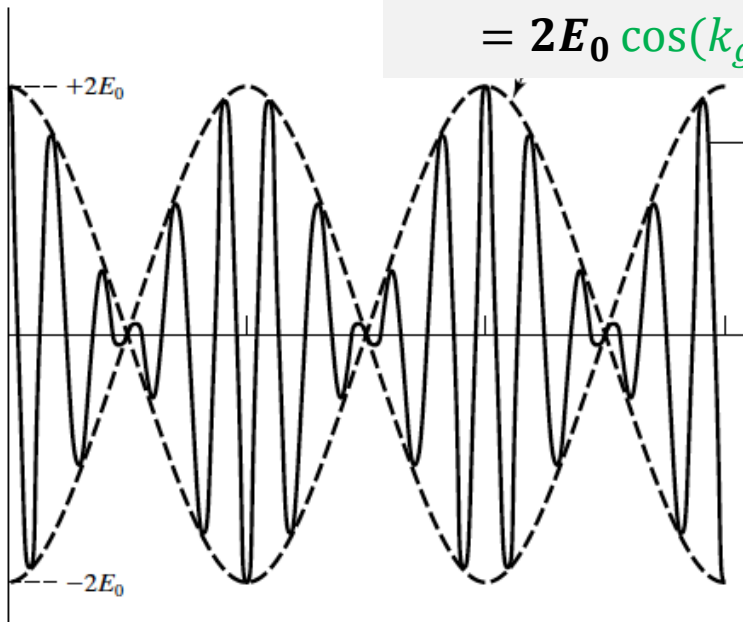
$$\lambda_m = \frac{2d}{m}$$

REVIEW-SUPERPOSITION (III)

Beating

$$\mathbf{E}_1(\mathbf{r}, t) = E_0 \cos(k_1 x - \omega_1 t) \quad \mathbf{E}_2(\mathbf{r}, t) = E_0 \cos(k_2 x - \omega_2 t)$$

$$\mathbf{E}_r(\mathbf{r}, t) = \mathbf{E}_1 + \mathbf{E}_2 = 2E_0 \cos\left(\frac{k_1 + k_2}{2}x - \frac{\omega_1 + \omega_2}{2}t\right) \cos\left(\frac{k_1 - k_2}{2}x - \frac{\omega_1 - \omega_2}{2}t\right)$$
$$= 2E_0 \cos(k_g t - \omega_g t) \cos(k_p t - \omega_p t)$$

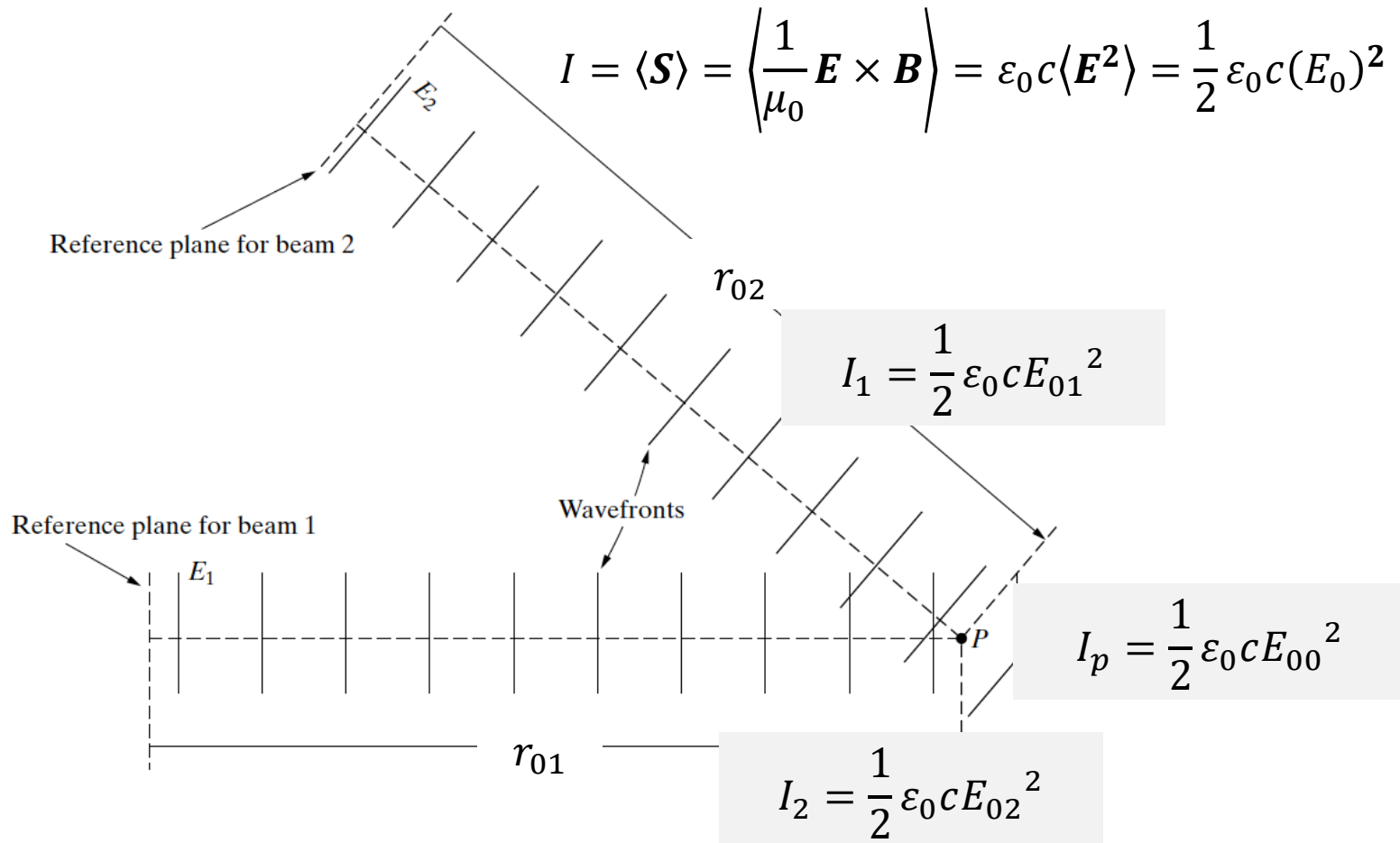


$$v_g = \frac{\omega_g}{k_g} \quad v_p = \frac{\omega_p}{k_p}$$

$$v_g \approx \frac{d\omega}{dk}$$

$$v_g = v_p \left(1 + \frac{\lambda}{n} \frac{dn}{d\lambda}\right)$$

4.3 INTERFERENCE OF LIGHT



4.3 INTERFERENCE OF LIGHT

$$\mathbf{E}_1(\mathbf{r}, t) = \mathbf{E}_{01} \sin(k_1 x_1 - \omega t)$$

$$\mathbf{E}_2(\mathbf{r}, t) = \mathbf{E}_{02} \sin(k_2 x_2 - \omega t)$$

$$\mathbf{E}_p(\mathbf{r}, t) = \mathbf{E}_1 + \mathbf{E}_2$$

$$I_p = \varepsilon_0 c \langle \mathbf{E}_p^2 \rangle = \varepsilon_0 c \langle \mathbf{E}_1^2 + \mathbf{E}_2^2 + 2\mathbf{E}_1 \cdot \mathbf{E}_2 \rangle = \varepsilon_0 c \langle \mathbf{E}_1^2 \rangle + \varepsilon_0 c \langle \mathbf{E}_2^2 \rangle + \varepsilon_0 c \langle 2\mathbf{E}_1 \cdot \mathbf{E}_2 \rangle$$

$$I_1 = \varepsilon_0 c \langle \mathbf{E}_1^2 \rangle = \frac{1}{2} \varepsilon_0 c E_{01}^2 \quad I_2 = \varepsilon_0 c \langle \mathbf{E}_2^2 \rangle = \frac{1}{2} \varepsilon_0 c E_{02}^2$$

$$I_{12} = 2\varepsilon_0 c \langle \mathbf{E}_1 \cdot \mathbf{E}_2 \rangle \quad \text{Interference term}$$

Wave nature of light

$$I = I_1 + I_2 + I_{12}$$

Polarization sensitive

4.3 INTERFERENCE OF LIGHT

$$\mathbf{E}_1(\mathbf{r}, t) = \mathbf{E}_{01} \sin(k \cdot \mathbf{r}_{01} - \omega t + \varphi_{01}) = \mathbf{E}_{01} \sin(\alpha_{01} - \omega t) \quad k \cdot \mathbf{r}_{01} + \varphi_{01} = \alpha_{01}$$

$$\mathbf{E}_2(\mathbf{r}, t) = \mathbf{E}_{02} \sin(k \cdot \mathbf{r}_{02} - \omega t + \varphi_{02}) = \mathbf{E}_{02} \sin(\alpha_{02} - \omega t) \quad k \cdot \mathbf{r}_{02} + \varphi_{02} = \alpha_{02}$$

$$\mathbf{E}_1 \cdot \mathbf{E}_2 = \mathbf{E}_{01} \cdot \mathbf{E}_{02} \sin(\alpha_{01} - \omega t) \sin(\alpha_{02} - \omega t)$$

$$= \frac{1}{2} \mathbf{E}_{01} \cdot \mathbf{E}_{02} [\cos(\alpha_{01} - \alpha_{02}) + \cos(\alpha_{01} + \alpha_{01} - 2\omega t)]$$

$$\langle \mathbf{E}_1 \cdot \mathbf{E}_2 \rangle = \frac{1}{2} \mathbf{E}_{01} \cdot \mathbf{E}_{02} [\cos(\alpha_{01} - \alpha_{02}) + 0]$$

$$I_{12} = 2\varepsilon_0 c \langle \mathbf{E}_1 \cdot \mathbf{E}_2 \rangle = 2\sqrt{I_1} \sqrt{I_2} \cos(\alpha_{01} - \alpha_{02})$$

$$I_{12} = 2\sqrt{I_1 I_2} \cos \delta$$

$$\delta = \alpha_{01} - \alpha_{02} = k \cdot \mathbf{r}_{01} + \varphi_{01} - k \cdot \mathbf{r}_{02} - \varphi_{02}$$

4.3 INTERFERENCE OF LIGHT

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos \delta$$

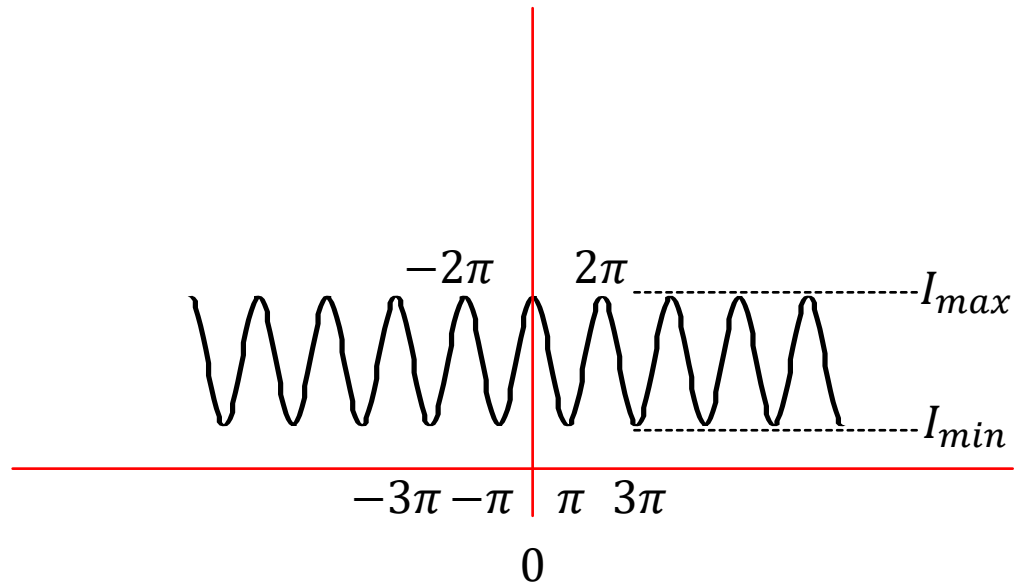
$$I_{max} = I_1 + I_2 + 2\sqrt{I_1 I_2}$$

$$I_{min} = I_1 + I_2 - 2\sqrt{I_1 I_2}$$

$$contrast = \frac{I_{max} - I_{min}}{I_{max} + I_{min}}$$

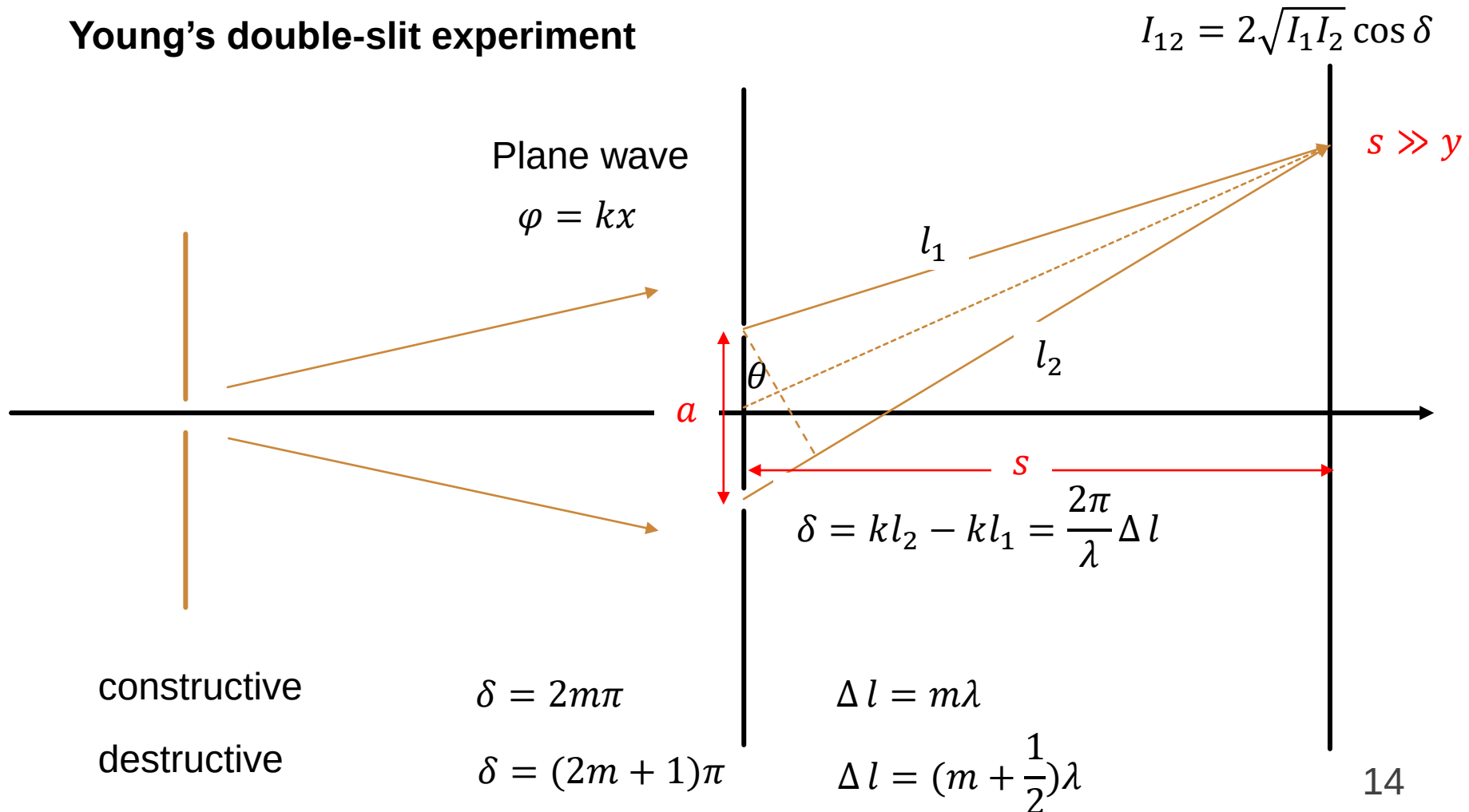
$$I_1 = I_2 = I_0$$

$$I_{max} = 4I_0 \quad I_{min} = 0$$



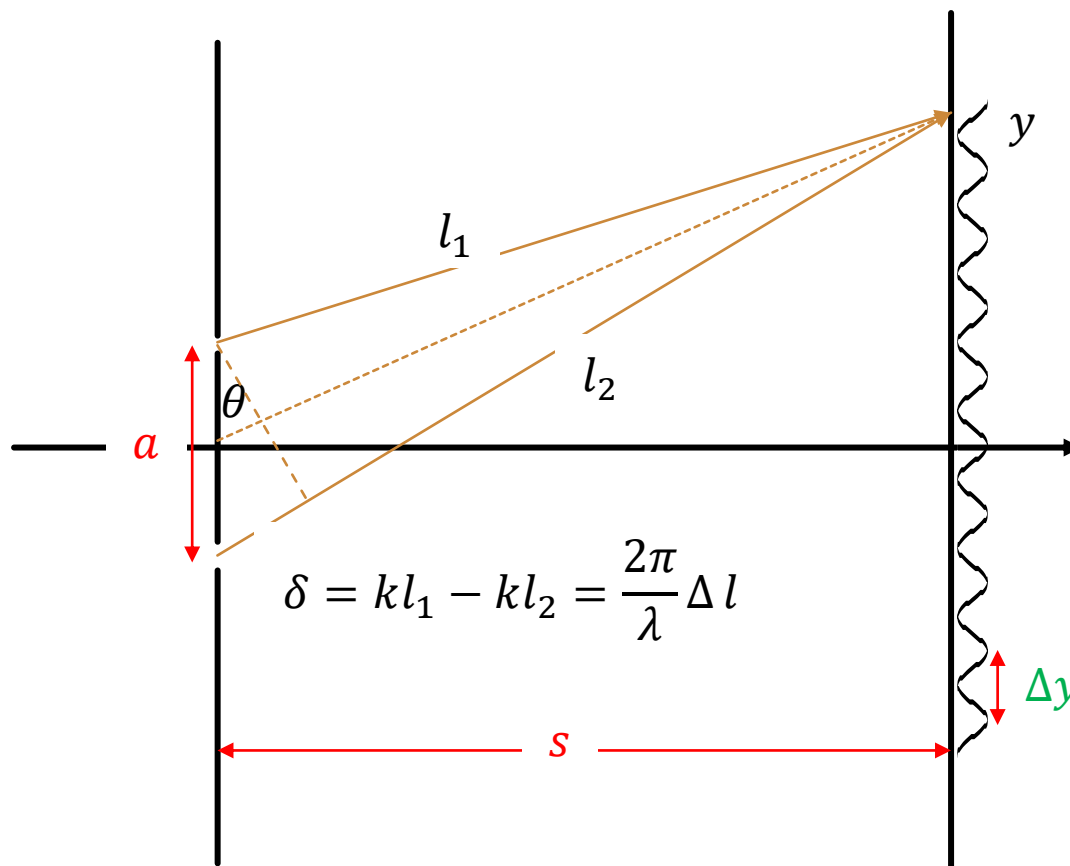
4.3 INTERFERENCE OF LIGHT

Young's double-slit experiment



4.3 INTERFERENCE OF LIGHT

Young's double-slit experiment



$$I = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos \delta$$

$$I = 2I_0 + 2I_0 \cos \delta$$

$$I = 2I_0(1 + \cos \delta) = 4I_0 \cos^2 \frac{\delta}{2}$$

$$I = 4I_0 \cos^2 \left(\frac{\pi}{\lambda} \Delta l \right)$$

$$s \gg y$$

$$I = 4I_0 \cos^2 \left(\frac{\pi a y}{\lambda s} \right)$$

$$\Delta y = \lambda s / a$$

4.3 INTERFERENCE OF LIGHT

Interference in thin films

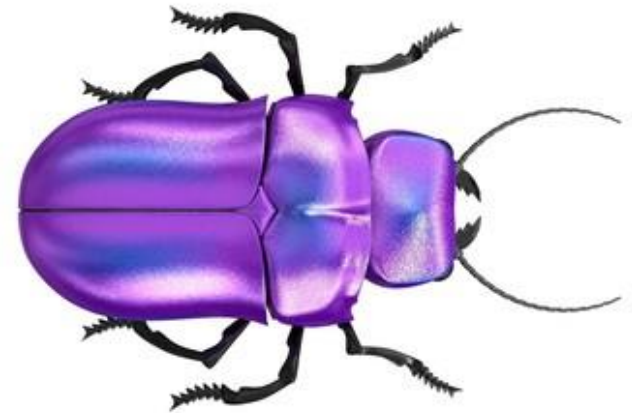
Dielectric mirrors



Soap bubbles

$$\Delta y = \lambda s/a$$

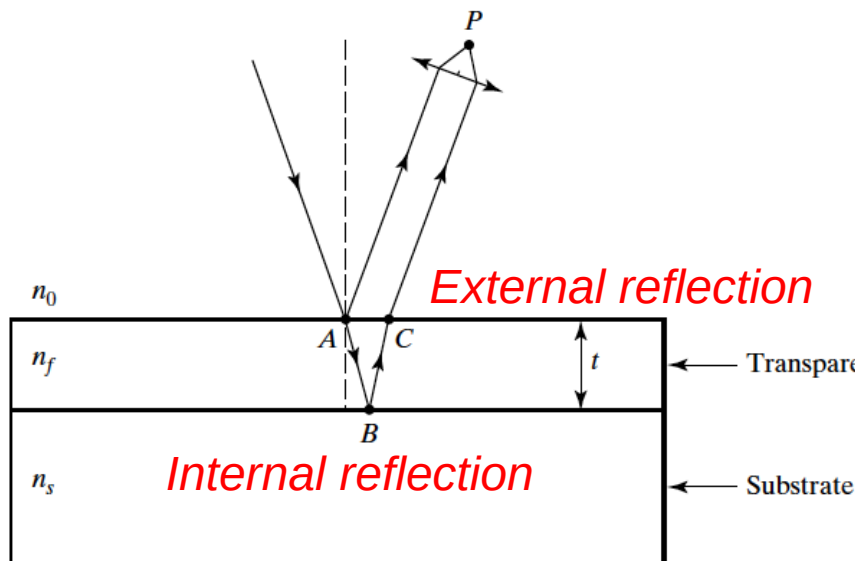
beetle shell



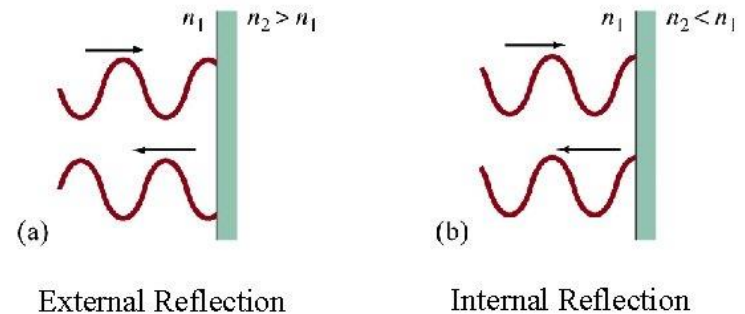
4.3 INTERFERENCE OF LIGHT

Interference in thin films

$$t \sim \lambda$$



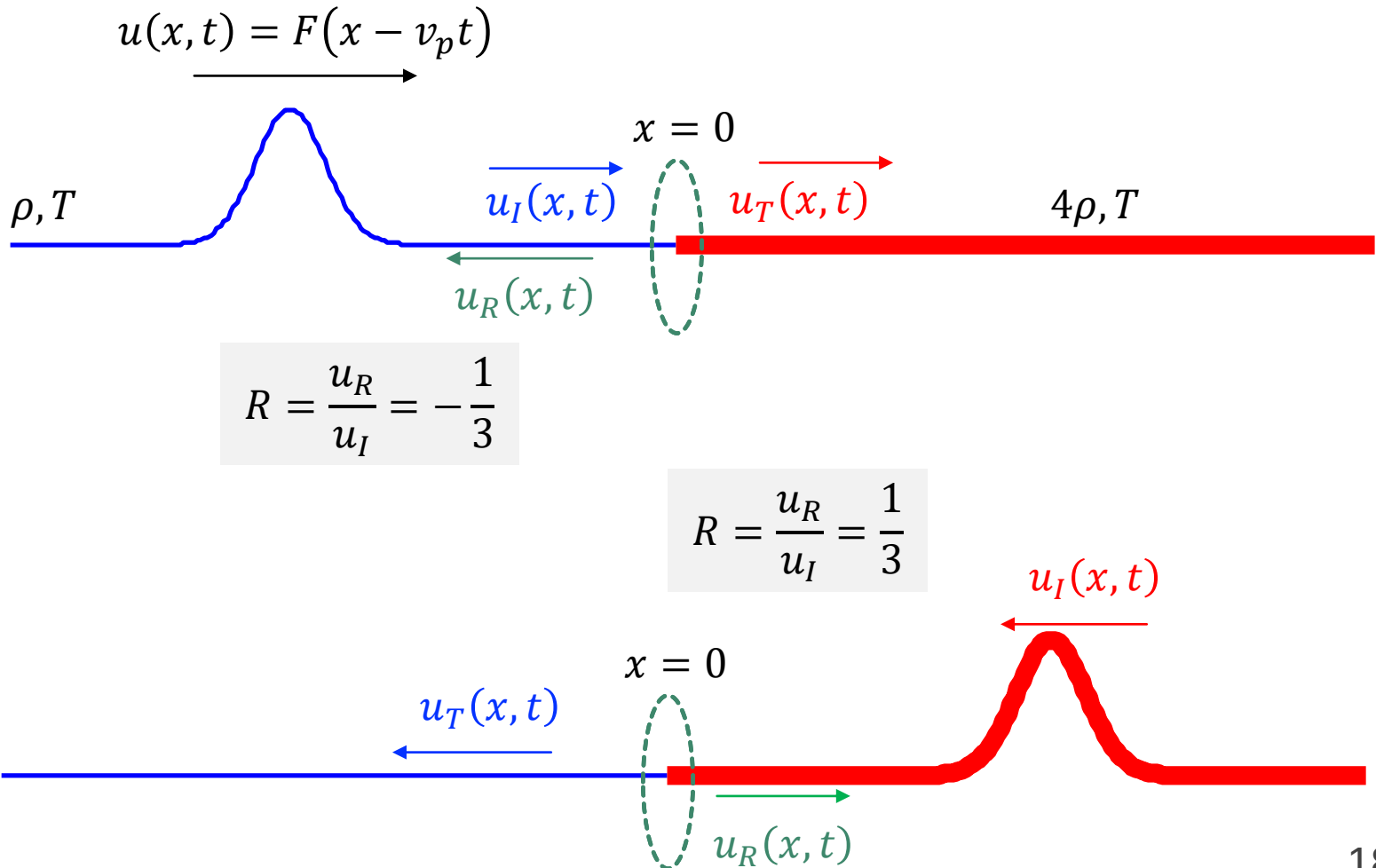
Phase shift : π



Now, If one reflection is internal and one reflection is external half wavelength path differences will result in constructive interference

CHAPTER 4. ELECTROMAGNETIC WAVES

REVIEW



4.3 INTERFERENCE OF LIGHT

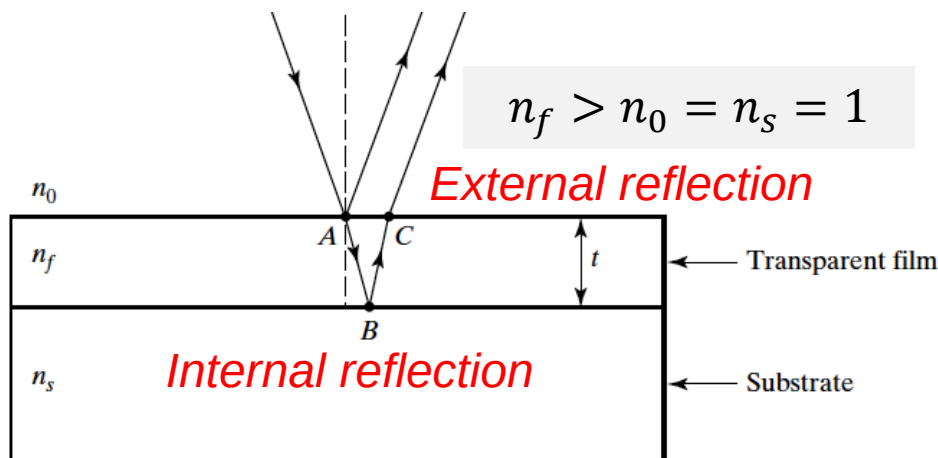
Interference in thin films

$$t \sim \lambda$$

Normal incidence:

$$\delta = kn(AB + BC) = 2knt$$

Phase shift : π



$$\delta = kn(AB + BC) + \pi = 2knt + \pi$$

$$\Delta = 2nt + \frac{1}{2}\lambda$$

constructive $\Delta = m\lambda$

destructive $\Delta = (m + \frac{1}{2})\lambda$

4.3 INTERFERENCE OF LIGHT

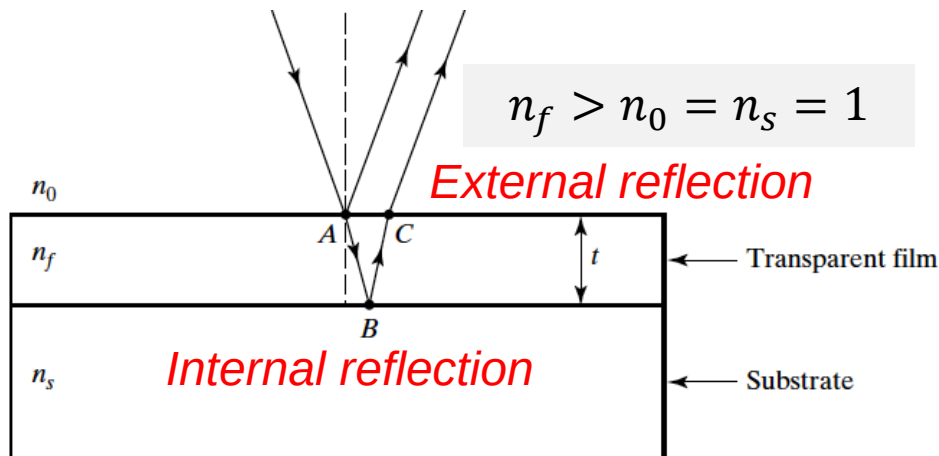
Interference in thin films

$$t \sim \lambda$$

Normal incidence:

$$\delta = kn(AB + BC) = 2knt$$

Phase shift : π



$$\delta = kn(AB + BC) + \pi = 2knt + \pi$$

$$\Delta = 2nt + \frac{1}{2}\lambda$$

constructive $nt = \frac{2m + 1}{4}\lambda$

destructive $nt = \frac{m}{2}\lambda$

4.3 INTERFERENCE OF LIGHT

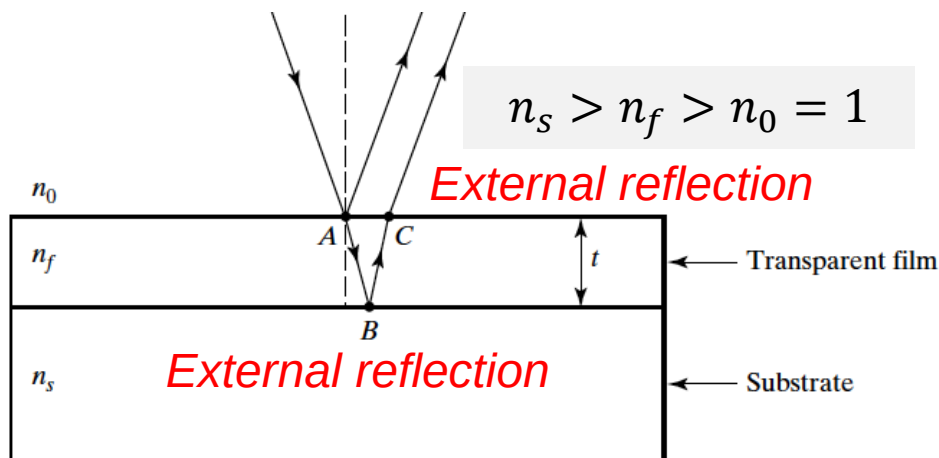
Interference in thin films

$$t \sim \lambda$$

Normal incidence:

$$\delta = kn(AB + BC) = 2knt$$

Phase shift : 0



$$\delta = kn(AB + BC) = 2knt$$

$$\Delta = 2nt$$

constructive $nt = \frac{m}{2}\lambda$

destructive $nt = \frac{2m + 1}{4}\lambda$

4.3 INTERFERENCE OF LIGHT

Interference in thin films



$$n_s > n_f > n_0 = 1$$

antireflecting coatings

destructive

$$\Delta = 2nt = \frac{1}{2}\lambda$$

$$nt = \frac{1}{4}\lambda$$

Highly reflective coatings

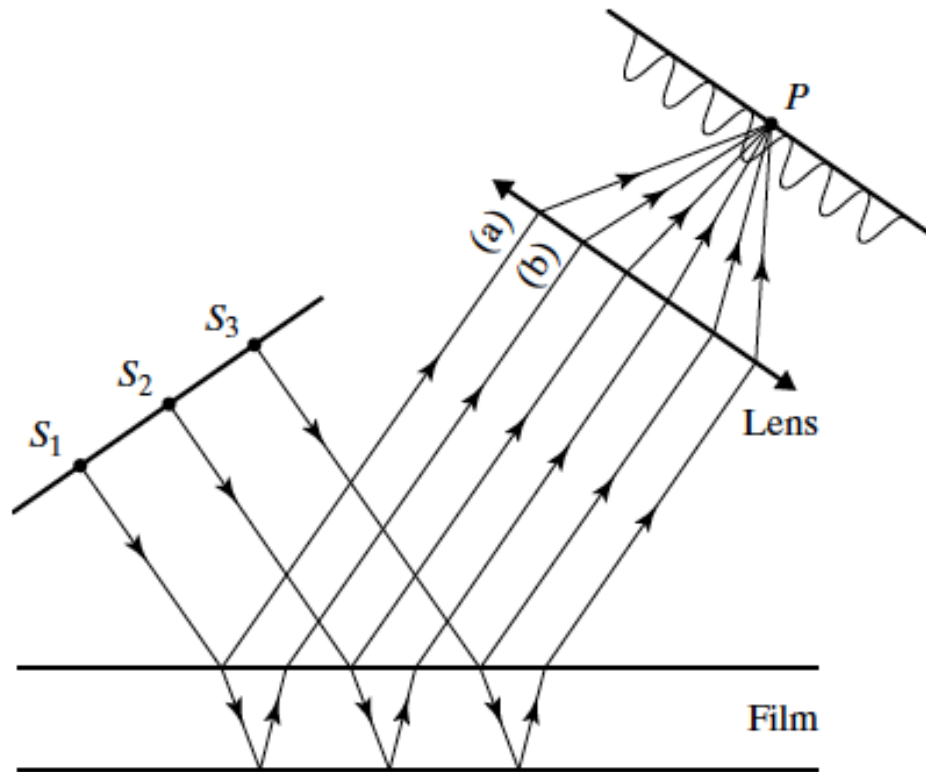
constructive

$$\Delta = 2nt = \lambda$$

$$nt = \frac{1}{2}\lambda$$

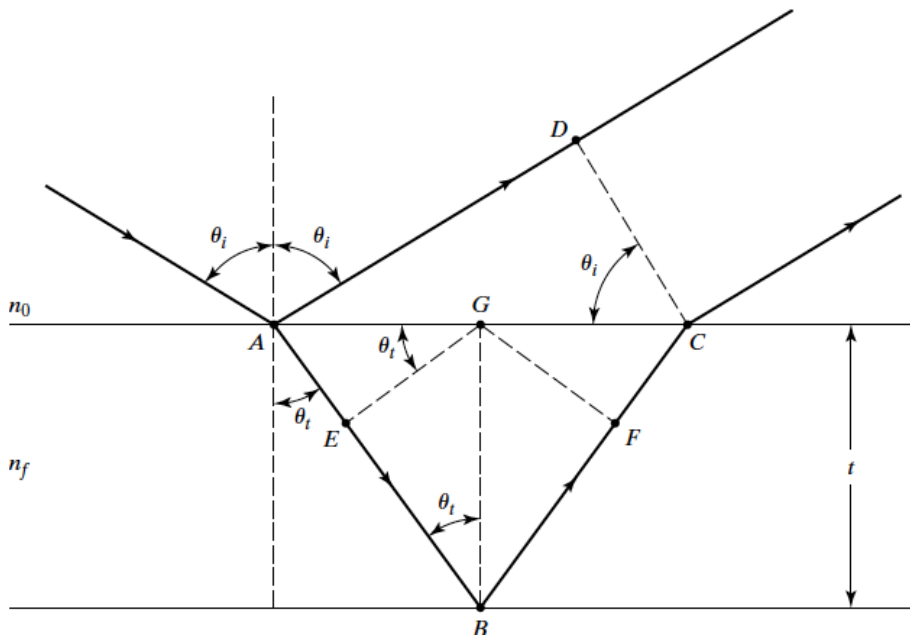
4.3 INTERFERENCE OF LIGHT

Interference in thin films (equal inclination)



4.3 INTERFERENCE OF LIGHT

Interference in thin films (equal inclination)



$$\Delta = n(AB + BC) - AD + \frac{1}{2}\lambda$$

$$AB = BC = t / \cos \theta_t$$

$$AD = 2t \tan \theta_t \sin \theta_i \quad \frac{\sin \theta_i}{\sin \theta_t} = n$$

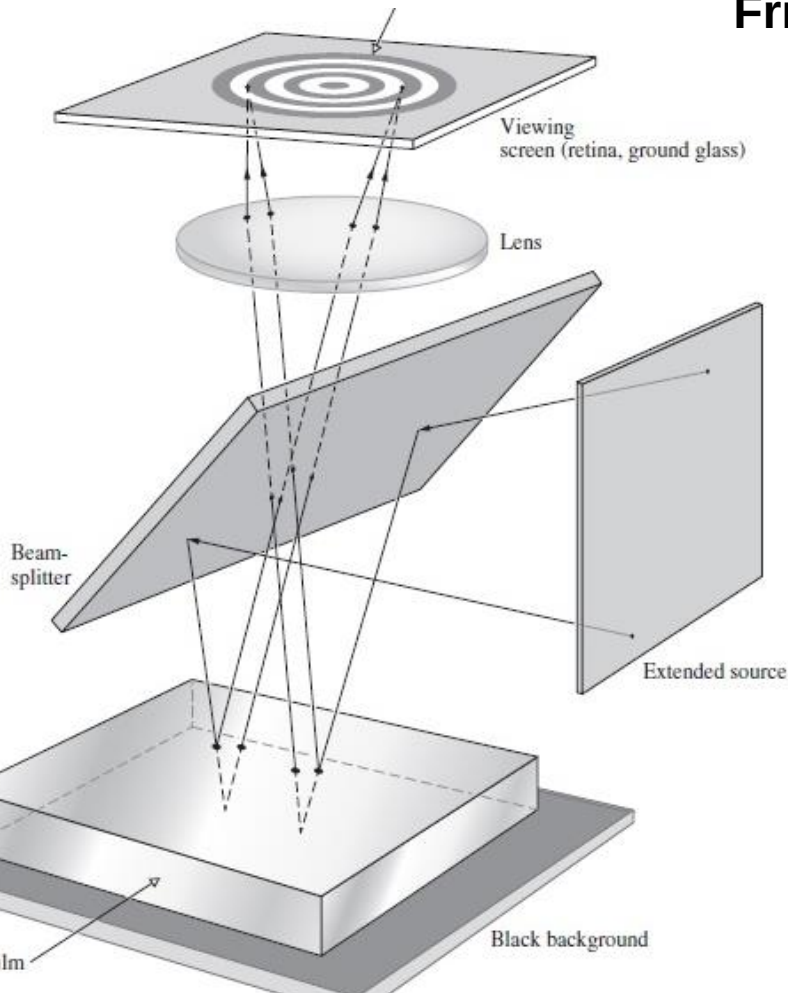
$$\Delta = 2nt \frac{1 - (\sin \theta_t)^2}{\cos \theta_t} + \frac{1}{2}\lambda$$

$$\Delta = 2t\sqrt{n^2 - \sin^2 \theta_i} + \frac{1}{2}\lambda$$

4.3 INTERFERENCE OF LIGHT

Constructive

Fringes of equal inclination



$$\Delta = 2t\sqrt{n^2 - \sin^2\theta_i} + \frac{1}{2}\lambda$$

Constructive

$$\Delta = 2t\sqrt{n^2 - \sin^2\theta_i} + \frac{1}{2}\lambda = m\lambda$$

Destructive

$$\Delta = 2t\sqrt{n^2 - \sin^2\theta_i} + \frac{1}{2}\lambda = \frac{2m+1}{2}\lambda$$

4.3 INTERFERENCE OF LIGHT

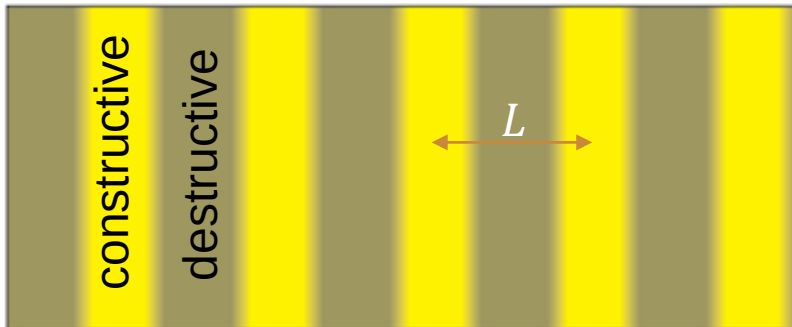
Interference from wedge-shaped films (equal thickness)



constructive $\Delta = 2nt + \frac{1}{2}\lambda = m\lambda$

destructive $\Delta = 2nt + \frac{1}{2}\lambda = \frac{2m+1}{2}\lambda$

Top view



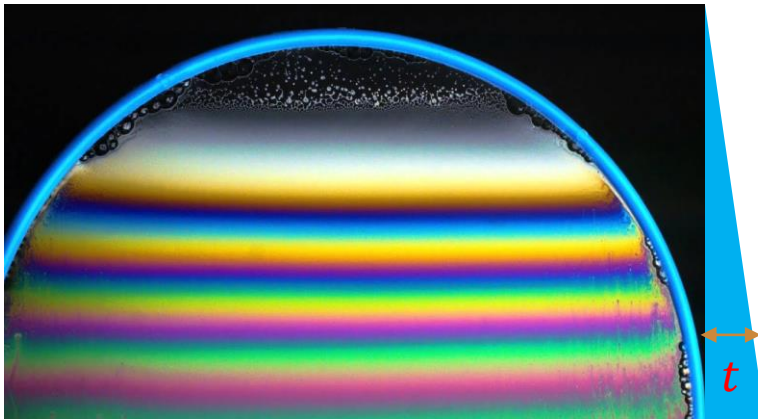
Fringe Interval:

$$L \alpha = (t_n - t_{n-1}) = \frac{\lambda}{2n}$$

$$L = \frac{\lambda}{2n\alpha}$$

4.3 INTERFERENCE OF LIGHT

Interference from wedge-shaped films (equal thickness)

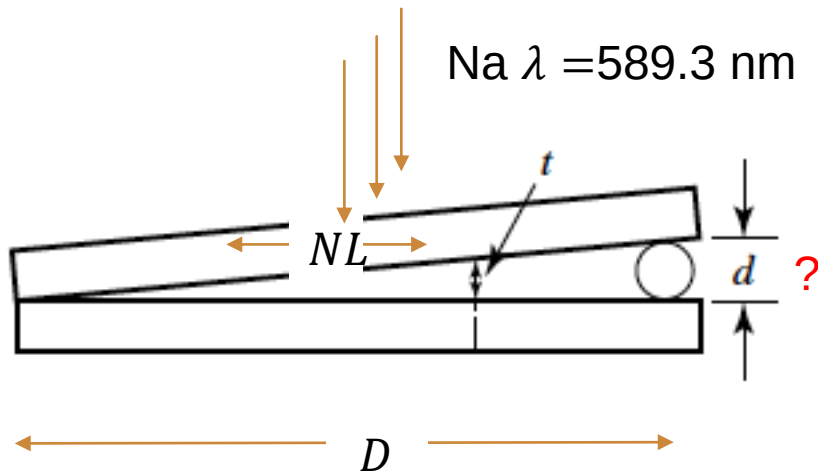


constructive $\Delta = 2nt_1 + \frac{1}{2}\lambda_1 = m\lambda_1$

$$\Delta = 2nt_2 + \frac{1}{2}\lambda_2 = m\lambda_2$$
$$\vdots$$

4.3 INTERFERENCE OF LIGHT

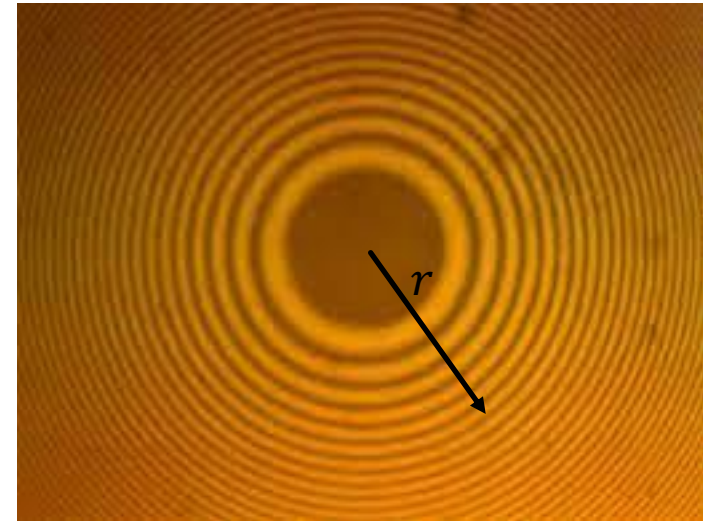
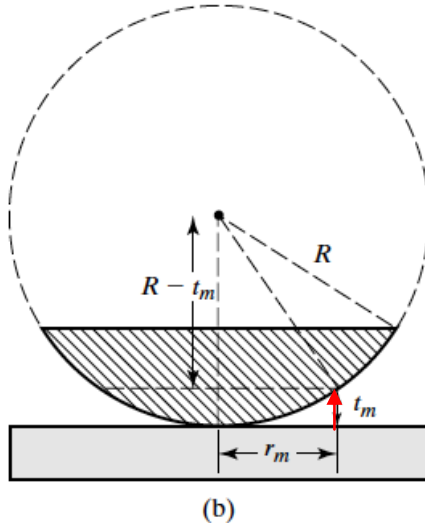
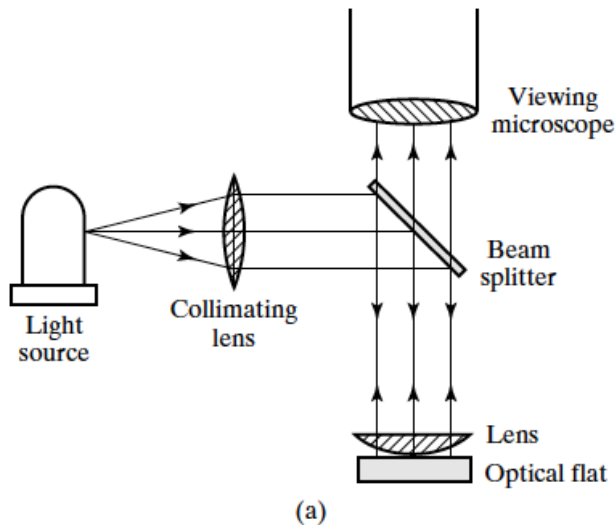
Interference from wedge-shaped films (equal thickness)



$$L = \frac{\lambda}{2n\alpha}$$

4.3 INTERFERENCE OF LIGHT

Newton's ring (equal thickness)



constructive $2t + \frac{1}{2}\lambda = m\lambda$

constructive

$$r^2 = R^2 - (R - t)^2 \approx 2Rt$$

$$r = \sqrt{\frac{2m-1}{2} R \lambda}$$

4.4 COHERENCE

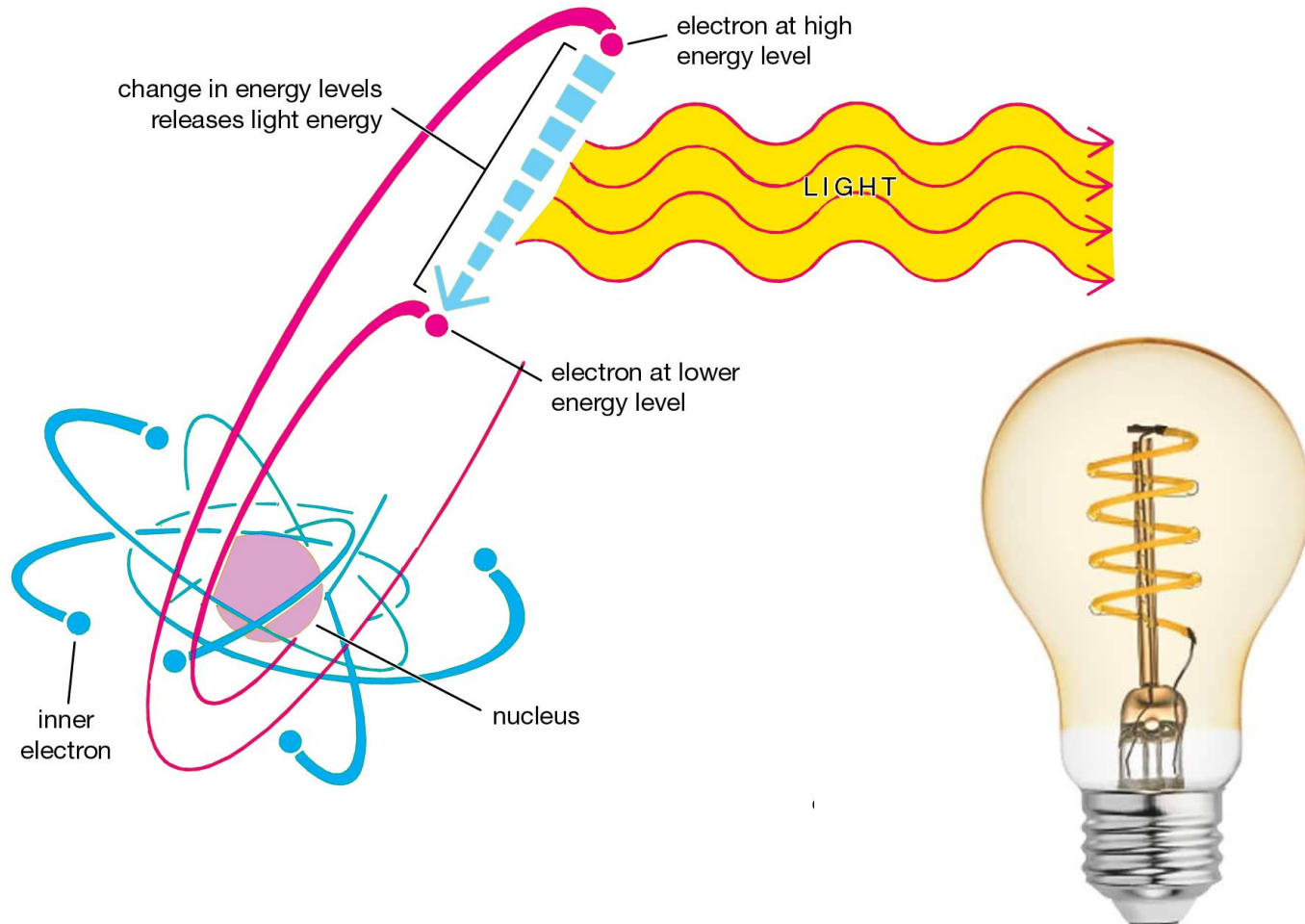
Interference term $I_{12} = 2\sqrt{I_1 I_2} \cos \delta$

$$\delta = \alpha_{01} - \alpha_{02} = \mathbf{k} \cdot \mathbf{r}_{01} + \varphi_{01} - \mathbf{k} \cdot \mathbf{r}_{02} - \varphi_{02}$$



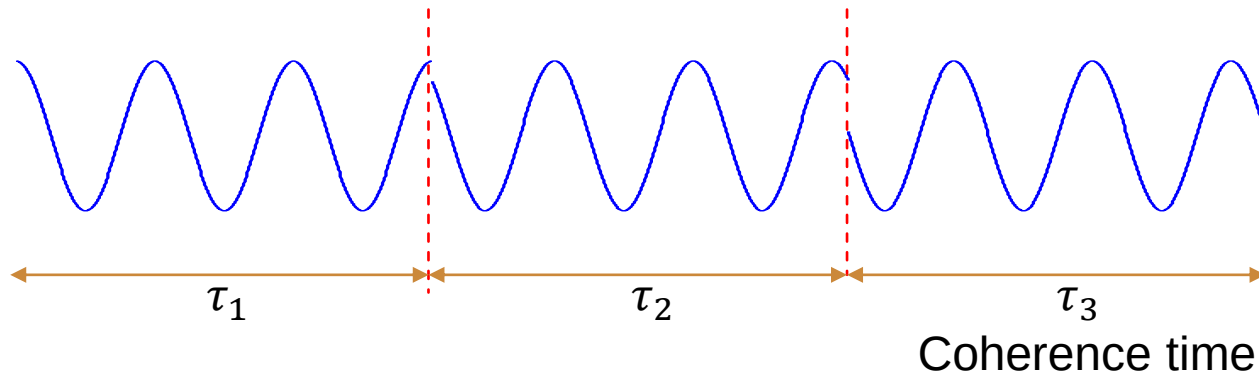
The term coherence is used to describe the correlation between phases of monochromatic radiations. Beams with random phase relationships are, generally speaking, incoherent beams, whereas beams with a constant phase relationship are coherent beams.

4.4 COHERENCE

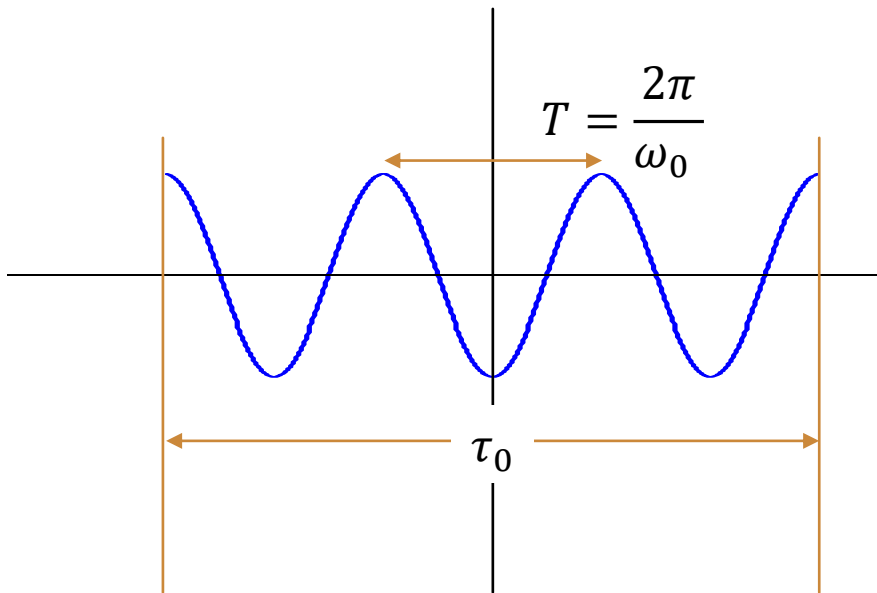


4.4 COHERENCE

“Monochromatic” light



4.4 COHERENCE



Radiation is on and off, finite length

$$f(t) = \begin{cases} -\cos(\omega_0 t), & -\frac{\tau_0}{2} < t < \frac{\tau_0}{2} \\ 0, & \text{elsewhere} \end{cases}$$

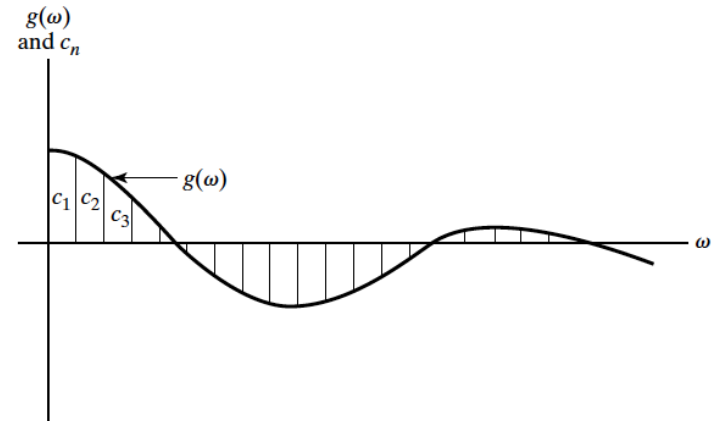
Fourier series:

$$f(t) = c_0 + \sum_{m=1, \dots} c_m \cos\left(\frac{2\pi}{\tau_0} m t\right)$$

$$\omega_m = \frac{2\pi}{\tau_0} m$$

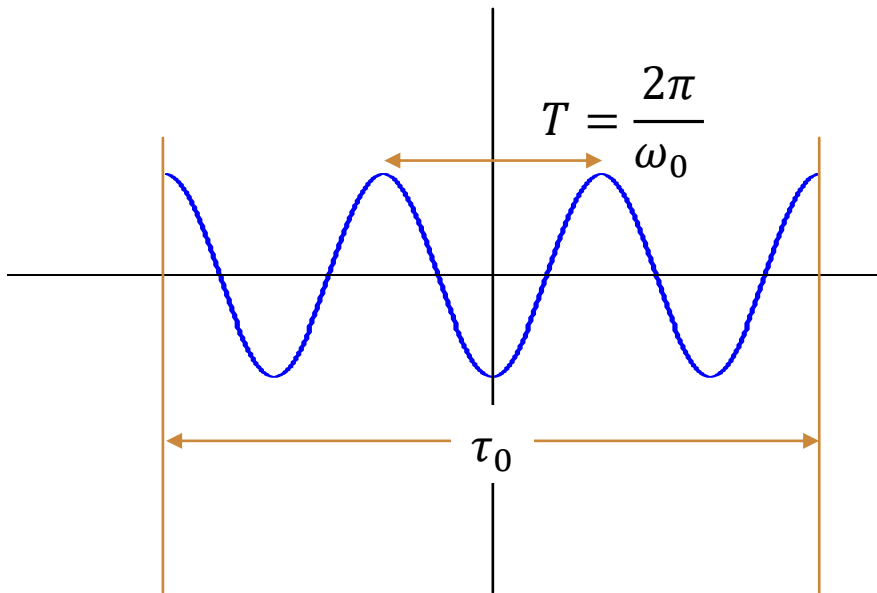
$$f(t) = c_0 + \sum_{m=1, \dots} c_m \cos(\omega_m t)$$

4.4 COHERENCE



$$f(t) = c_0 + \sum_{m=1,\dots} c_m \cos(\omega_m t)$$

4.4 COHERENCE



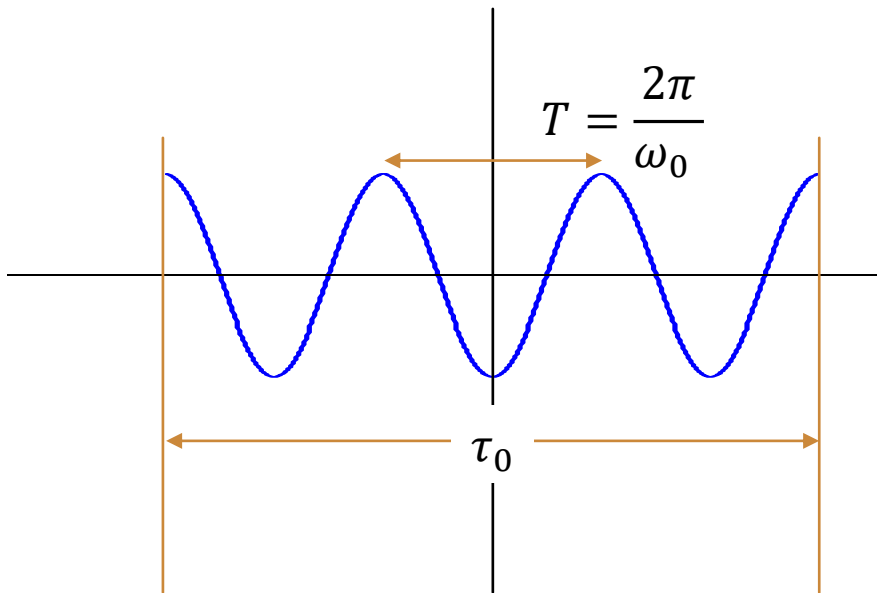
Radiation is on and off, finite length

$$f(t) = \begin{cases} e^{-i\omega_0 t}, & -\frac{\tau_0}{2} < t < \frac{\tau_0}{2} \\ 0, & \text{elsewhere} \end{cases}$$

Fourier transform

$$f(t) = \int_{-\infty}^{+\infty} g(\omega) e^{-i\omega t} d\omega$$

4.4 COHERENCE



Radiation is on and off, finite length

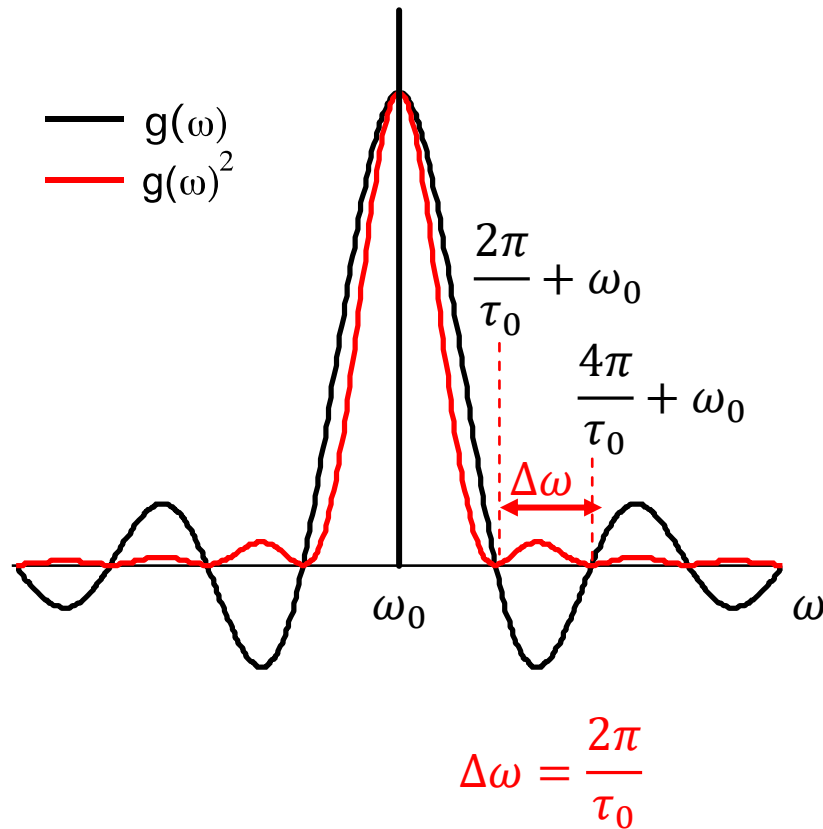
$$f(t) = \begin{cases} e^{-i\omega_0 t}, & -\frac{\tau_0}{2} < t < \frac{\tau_0}{2} \\ 0, & \text{elsewhere} \end{cases}$$

Fourier transform

$$g(\omega) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} f(t) e^{i\omega t} dt$$

$$\begin{aligned} g(\omega) &= \frac{1}{2\pi} \int_{-\frac{\tau_0}{2}}^{+\frac{\tau_0}{2}} e^{i(\omega - \omega_0)t} dt = \frac{1}{\pi(\omega - \omega_0)} \left[\frac{e^{i(\omega - \omega_0)\frac{\tau_0}{2}} - e^{-i(\omega - \omega_0)\frac{\tau_0}{2}}}{2i} \right] = \frac{\sin \frac{(\omega - \omega_0)\tau_0}{2}}{\pi(\omega - \omega_0)} \\ &= \frac{\tau_0}{2\pi} \frac{\sin u}{u} \end{aligned}$$

4.4 COHERENCE



$$g(\omega) = \frac{\tau_0}{2\pi} \frac{\sin u}{u} \quad u = \frac{(\omega - \omega_0)\tau_0}{2}$$

$$\lim_{\omega \rightarrow \omega_0} g(\omega) = \frac{\tau_0}{2\pi}$$

$$g(\omega) = 0, \quad \frac{(\omega - \omega_0)\tau_0}{2} = n\pi$$

$$\omega = \frac{2n\pi}{\tau_0} + \omega_0$$

the shorter τ_0 , the wider $g(\omega)$

$$\tau_0 \rightarrow \infty, \quad \omega \rightarrow \omega_0$$

4.4 COHERENCE

$$\Delta\omega = \frac{2\pi}{\tau_i}$$

The natural width of a spectral line is inversely proportional to the coherence time of the source

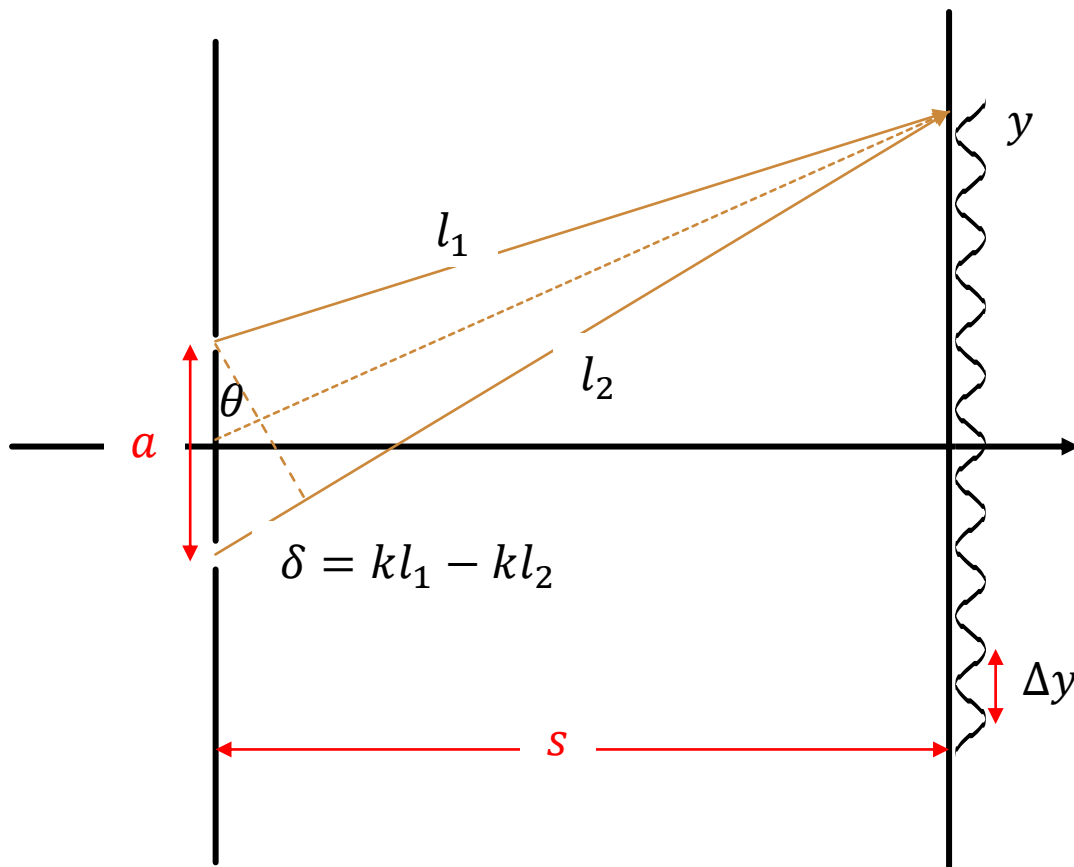
$$\Delta\omega = 2\pi \frac{c}{\lambda^2} \Delta\lambda$$


$$\left[\omega = 2\pi \frac{c}{\lambda} \right]$$

$$\Delta\lambda = \frac{\Delta\omega \lambda^2}{2\pi c}$$

4.4 COHERENCE

Young's double-slit experiment



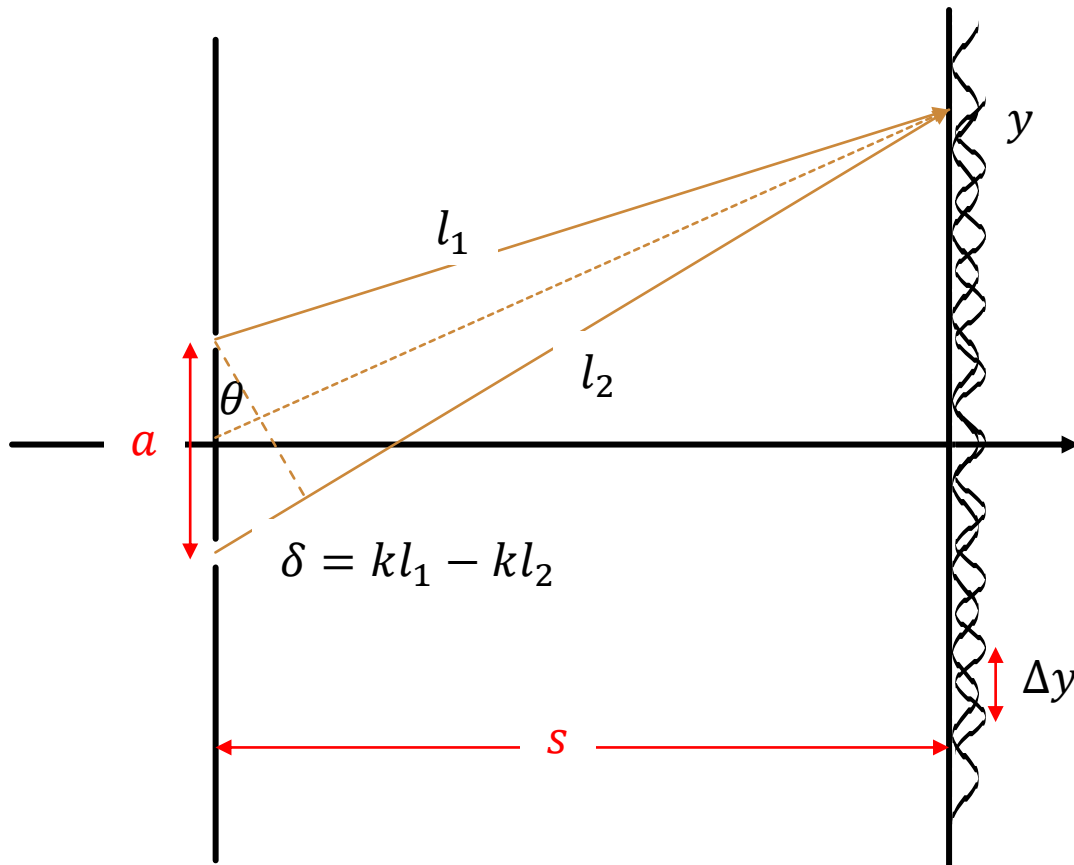
$$I = 4I_0 \cos^2 \left(\frac{\pi a y}{\lambda s} \right)$$

$$y = m \lambda s / a$$

$$\Delta y = \lambda s / a$$

4.4 COHERENCE

Young's double-slit experiment



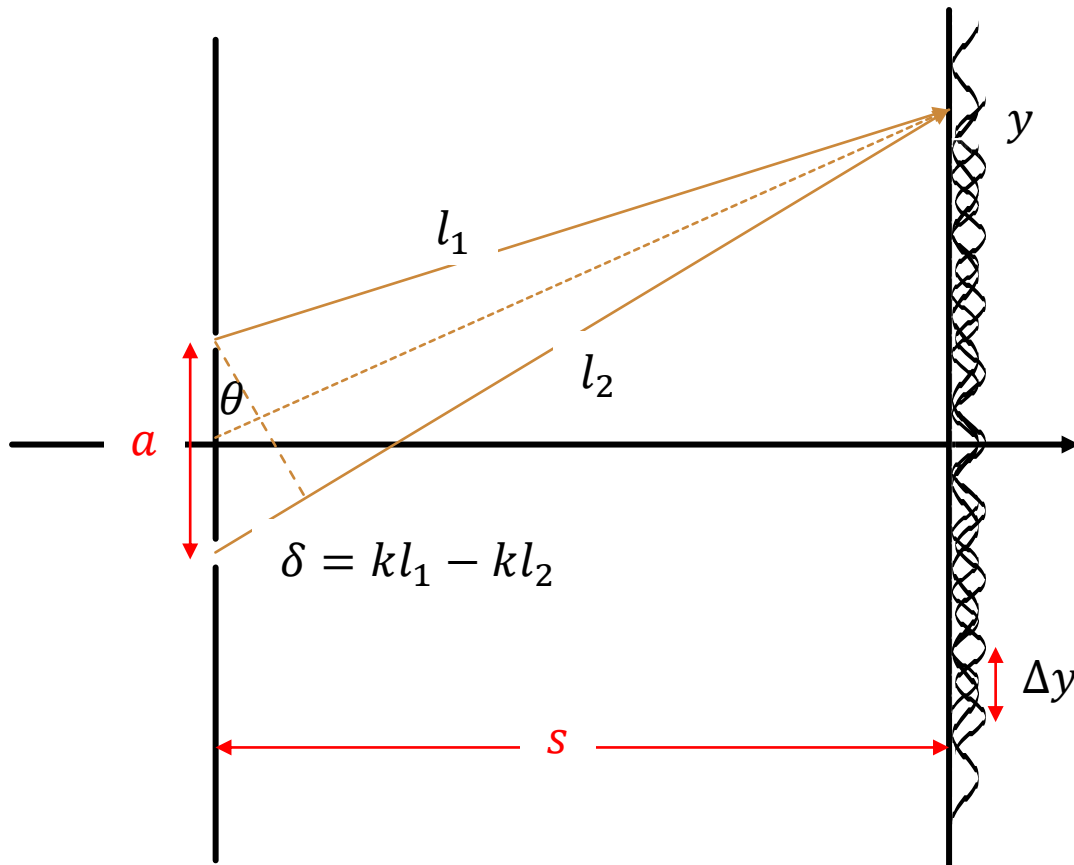
$$I = 4I_0 \cos^2 \left(\frac{\pi a y}{\lambda s} \right)$$

$$y = m \lambda s / a$$

$$\Delta y = \left(\lambda + \frac{\Delta \lambda}{2} \right) s / a$$

4.4 COHERENCE

Young's double-slit experiment



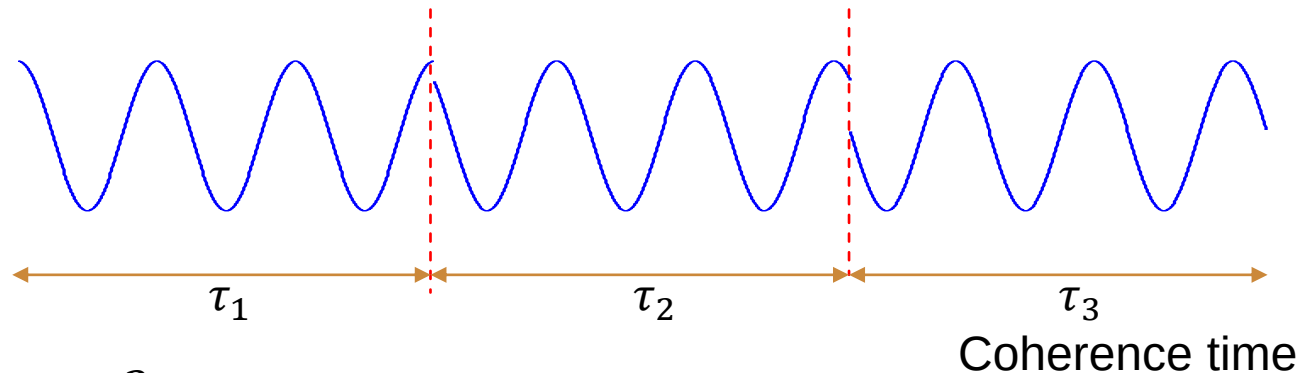
$$I = 4I_0 \cos^2 \left(\frac{\pi a y}{\lambda s} \right)$$

$$y = m \lambda s / a$$

$$\Delta y = \left(\lambda - \frac{\Delta \lambda}{2} \right) s / a$$

4.4 COHERENCE

“Monochromatic” light



$$\Delta\omega = \frac{2\pi}{\tau_i}$$

The natural width of a spectral line is inversely proportional to the coherence time of the source

$$\Delta\omega = 2\pi \frac{c}{\lambda^2} \Delta\lambda$$

Coherence length: $l_i = c\tau_i$ $l_i = c \frac{2\pi}{\Delta\omega} = \frac{\lambda^2}{\Delta\lambda}$

4.4 COHERENCE

“Monochromatic” light

$$l_i = \frac{\lambda^2}{\Delta\lambda}$$

Natural white-light

$$\lambda \sim 400 - 700nm$$
$$\Delta\lambda \sim 300nm$$

$$l_i \sim 1000nm$$

Mercury lamp

$$\lambda \sim 546nm$$
$$\Delta\lambda \sim 0.025nm$$

$$l_i \sim 1.2cm$$

CO₂ laser

$$\lambda \sim 10.6\mu m$$
$$\Delta\lambda \sim 10^{-5}nm$$

$$l_i \sim 11km$$

Fiber laser

$$l_i \sim 100km$$

Hydrogen maser

$$l_i \sim 300000km$$